

Week 1 – Bits & Bytes

Student number: 583259

Assignment 1.1: Bits & Bytes intro

What are Bits & Bytes?

- A bit is the smallest unit of data. It can be either 0 or 1.
- A byte is made up of 8 bits and can represent 256 different values.

What is a nibble?

- A *nibble* is a group of four bits, which is **half of a byte**. It can represent 16 possible values.

What relationship does a nibble have with a hexadecimal value?

- A nibble is made up of four bits, forming 16 possible combinations. This fits with the 16 symbols used in the hexadecimal system. It uses 16 symbols, 0 to 9 and A to F. Each hexadecimal digit is exactly one nibble, making a hexadecimal a simpler and readable way to express binary.

Why is it wise to display binary data as hexadecimal values?

- When hexadecimal values turn into binary numbers, they become long strings of 0s and 1s that's hard to read, and easy to make mistakes in. Using hexadecimal shortens them, each hexadecimal digit replaces four binary digits, or one **nibble**.

What kind of relationship does a byte have with a hexadecimal value?

- A byte is made up of eight bits, made up of two nibbles or of four bits. One byte equals two hexadecimal's digits.

An IPv4 subnet is 32-bit, show with a calculation why this is the case.

- An IPv4 address is made up of 4bytes, and each byte consists of 8 bits. When you multiply these, you get: $4 \text{ bytes} \times 8 \text{ bits per byte} = 32 \text{ bits}$. This means that every IPv4 address is 32 bits long.

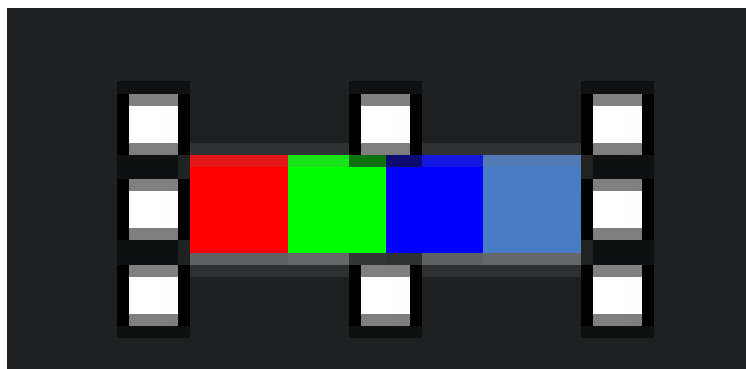
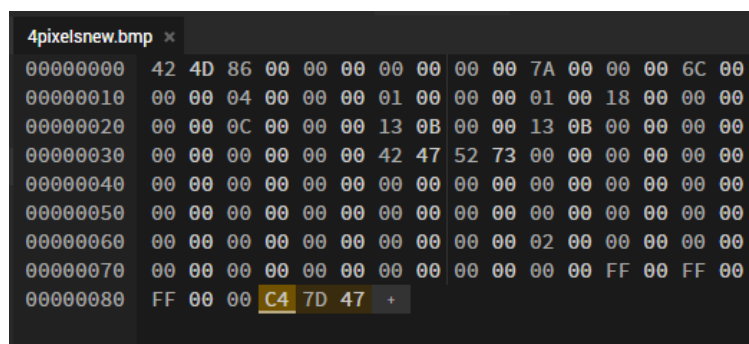
Assignment 1.2: Your favourite color

Hexadecimal color code: #477dc4

Assignment 1.3: Manipulating binary data

Color	Color code hexadecimal (RGB)	Big Endian	Little Endian
RED	#ff0000	ff0000	0000ff
GREEN	#00ff00	00ff00	00ff00
BLUE	#0000ff	0000ff	ff0000
WHITE	#ffffff	ffffff	ffffff
Favourite (previous assignment)	#477dc4	477dc4	c47d47

Screenshot modified BMP file in hex editor:



Assignment 1.4: Student number to HEX and Binary

Convert your student number to a hexadecimal number and a binary number.

Explain in detail that the calculation is correct. Use the PowerPoint slides of week 1.

Hexadecimal

- $583259 / 16 = 36453$ remainder is 11 -> B
- $36453 / 16 = 2278$ remainder is 5
- $2278 / 16 = 142$ remainder is 6
- $142 / 16 = 8$ remainder is 14 -> E
- $8 / 16 = 0$ remainder is 8

583259 (decimal) = 8E65B

Binary

- 8 -> 1000
- E -> 1110
- 6 -> 0110
- 5 -> 0101
- B -> 1011

583259 (decimal) = 8E65B (hexadecimal)

-> 1000 1110 0110 0101 1011 in binary.

Hex Digit	Binary Equivalent
0	0000
1	0001
2	0010
3	0011
4	0100
5	0101
6	0110
7	0111
8	1000
9	1001
A	1010
B	1011
C	1100
D	1101
E	1110
F	1111